

BEFORE

Throughout the late medieval period, physicians and “natural philosophers” began to develop what was later known as the “scientific method.”

ANATOMISTS

Taboos surrounding the desecration of corpses prevented physicians of the 2nd–14th centuries from studying human anatomy in any detail. As a result, the work of classical writers such as **Galen** and **Hippocrates** << 104–05 went unchallenged for centuries. It was only with the arrival of the **Black Death** << 186–87 in the late 1340s that the restrictions on dissection were eased. It took the genius of **Leonardo da Vinci** << 254–55 to turn anatomy into a truly scientific discipline.



CUTAWAY OF A MAN

ALCHEMISTS

A mix of **proto-science** and **superstition**, alchemy was the forerunner of chemistry that developed from the 11th century into the Renaissance (1300–1650 CE) << 250–53. Practitioners used methods we would recognize

today in pursuit of a mythical “philosopher’s stone” attributed with a variety of powers, such as the ability to give everlasting life or to make gold.



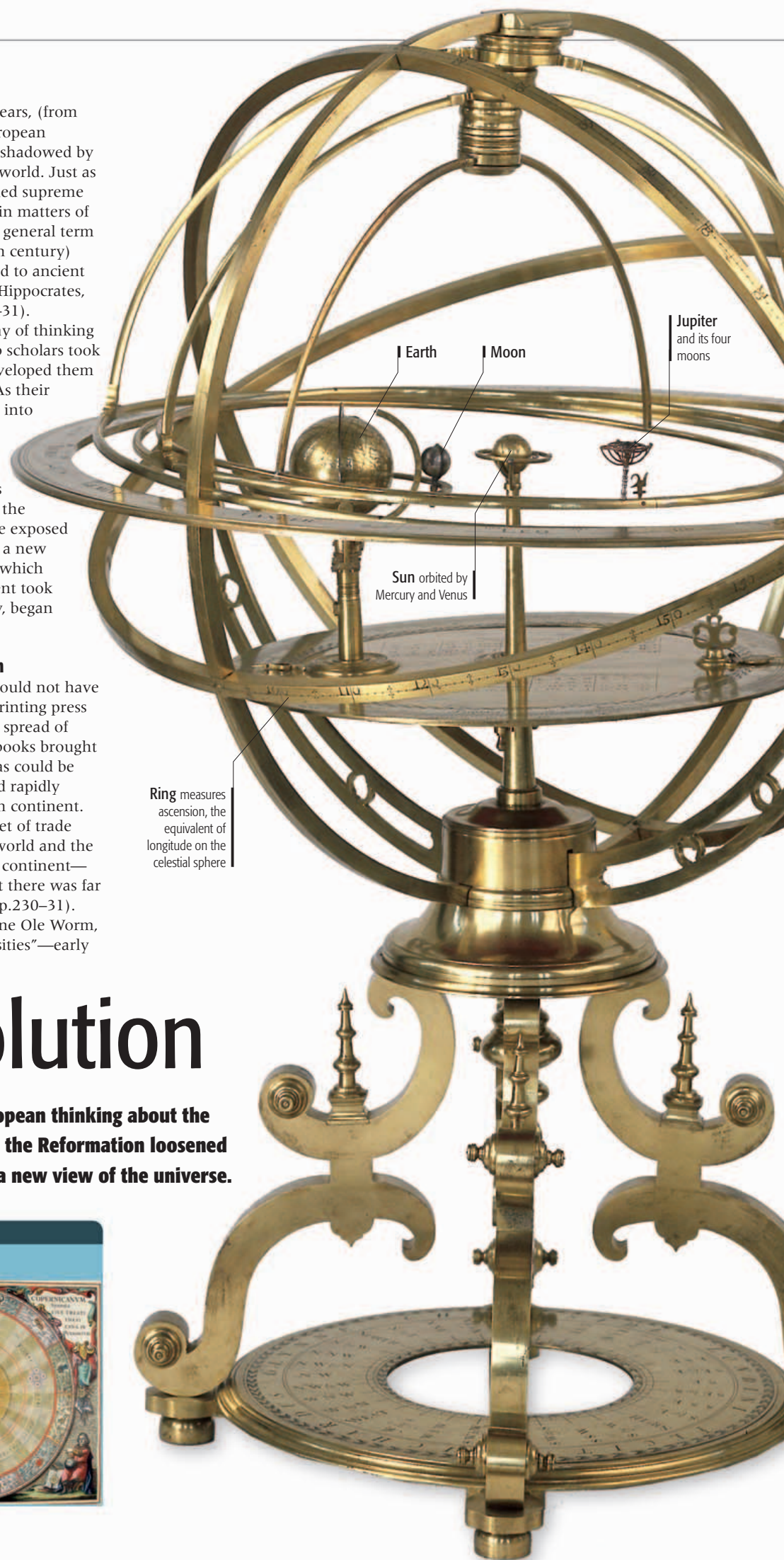
ALCHEMISTS AT WORK

For almost 1,500 years, (from about 200 CE) European thought was overshadowed by the legacy of the classical world. Just as the Catholic Church reigned supreme in religious affairs, so too in matters of “natural philosophy” (the general term for science before the 19th century) nearly all thinkers deferred to ancient authorities such as Plato, Hippocrates, and Aristotle (see pp.130–31).

Despite this, the new way of thinking had a long gestation. Arab scholars took many Greek ideas and developed them further (see pp.174–77). As their works began to filter back into European libraries, they helped to inspire a growing recognition that not everything the Greeks said was accurate. During the 18th century, failings were exposed in a number of areas, and a new method of philosophy, in which observation and experiment took precedence over authority, began to establish itself.

The knowledge explosion

The scientific revolution could not have taken place without the printing press (see p.253), and the rapid spread of information that printed books brought with them. Suddenly, ideas could be transmitted accurately and rapidly across the entire European continent. At the same time, the onset of trade with remote parts of the world and the discovery of a whole new continent—the Americas—meant that there was far more to investigate (see pp.230–31). Collectors, such as the Dane Ole Worm, created “cabinets of curiosities”—early



Scientific Revolution

The 16th and 17th centuries saw a metamorphosis in European thinking about the natural world. The Renaissance had transformed art, and the Reformation loosened the ties of religious dogma. A third revolution produced a new view of the universe.

DISCOVERY

THE HELIOCENTRIC UNIVERSE

The idea that the universe orbited Earth is an ancient one. Celestial objects were thought to be mounted on spheres. However, there were problems—Mercury and Venus never strayed far from the Sun, and Mars occasionally reversed its motion. In 1543, Polish priest and astronomer Nicolaus Copernicus published the first widely read proposal for a new system with the Sun at its center (pictured right), and the Earth the third of six planets orbiting around it.

